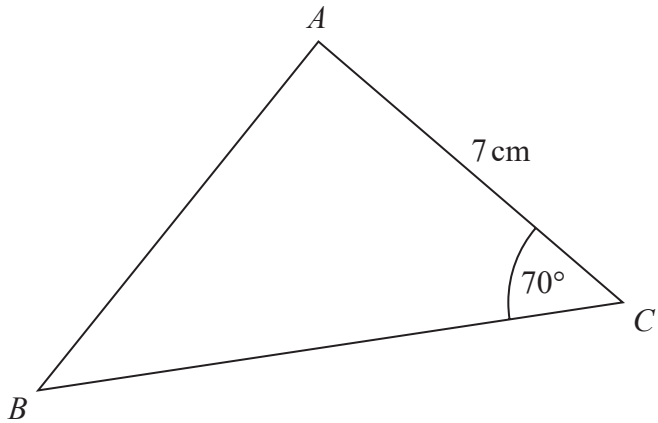
24



The area of triangle ABC is $42\,cm^2$

Find the length of AB.
Give your answer correct to 3 significant figures.

Use area = ½ × AC × BC × sin C.

42 = ½ × 7 × BC × sin70°

1 mark
Area substitution

BC = 84 ÷ (7sin70°) = 12.7701332697...cm.

1 mark
Find BC

Cosine rule: $AB^2 = 7^2 + BC^2 - 2 \times 7 \times BC \times \cos 70°$

1 mark
Cosine rule

= 150.929304369...

1 mark
Correct order

AB = √150.929304369... = 12.285328826...cm

= 12.3 cm (3 significant figures).

1 mark
12.3cm

..... cm

(Total for Question 24 is 5 marks)